CERTIFICATE OF ACCREDITATION

ANSI National Accreditation Board

11617 Coldwater Road, Fort Wayne, IN 46845 USA

This is to certify that

Micro Precision Calibration, Inc.

22835 Industrial Place
Grass Valley, CA 95949

has been assessed by ANAB and meets the requirements of international standard

ISO/IEC 17025:2017

and national standards

ANSI/NCSL Z540-1-1994 (R2002) and
ANSI/NCSL Z540.3-2006 (R2013)

while demonstrating technical competence in the field of

CALIBRATION

Refer to the accompanying Scope of Accreditation for information regarding the types of activities to which this accreditation applies

AC-1969.00
Certificate Number

ANAB Approval

Certificate Valid Through: 10/31/2020
Version No. 006 Issued: 06/04/2019

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated April 2017).

ANSI National Accreditation Board

SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017, ANSI/NCSL Z540-1-1994 (R2002) AND ANSI/NCSL Z540.3-2006 (R2013)

Micro Precision Calibration, Inc.

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CALIBRATION

Valid to: **October 31, 2020** Certificate Number: **AC-1969.00**

Acoustics and Vibration

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Sound[1] (125 to 2 000) Hz (2 000 to 4 000) Hz	 (74 to 114) dB SPL (74 to 114) dB SPL	 0.34 dB 0.45 dB	Sound calibrator Sound meter Electronic counter
Accelerometers	(1 to 5) g @ (10 to 2 500) Hz (2.5 to 10) kHz	 2.4 % of reading 2.9 % of reading	Vibration Calibration System

Chemical Quantities

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Conductivity Meters[1,3]	5.00 µS/cm 10.00 µS/cm 100.00 µS/cm 1 500 µS/cm 100 000 µS/cm	0.56 µS/cm 0.56 µS/cm 2.1 µS/cm 4.8 µS/cm 370 µS/cm	Standard Conductivity Solutions
pH Meters[1,3]	(4, 7, 10) pH units	0.02 pH unit	Standard pH Solutions
Calibration of Oxygen Analyzers	(0.1 to 1 000) parts in 10^6	3 % of reading	Standard gases
Optical System – Aerosol Particle Counter	Particle size: (µm) (0.1, 0.3, 0.5, 1, 2, 5, 10)	6 % of reading	Comparison to Standard Particle Counter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Voltage Source[1]	Up to 220 mV 220 mV to 2.2 V (2.2 to 11) V (11 to 22) V (22 to 220) V 220 V to 1.1 kV	7 µV/V + 0.4 µV 4 µV/V + 0.7 µV 3 µV/V + 2.5 µV 3 µV/V + 4 µV 4 µV/V + 40 µV 5.5 µV/V + 400 µV	Multiproduct Calibrator
DC Voltage Source[1] Fixed Values	1 V 1.018V 10 V	1.6 µV/V 1.6 µV/V 1.2 µV/V	DC Voltage Standard
DC Voltage Measure[1]	Up to 100 mV (0.1 to 1) V (1 to 10) V (10 to 100) V 100 V to 1 kV	3 µV/V + 0.3 µV 1.8 µV/V + 0.3 µV 0.6 µV/V + 0.5 µV 3 µV/V + 30 µV 3.6 µV/V + 0.1 mV	Multimeter
DC Voltage Measure	Up to 100 mV (0.1 to 1) V (1 to 10) V (10 to 100) V 100 V to 1 kV	1.6 µV/V 1.5 µV/V 1.5 µV/V 1.7 µV/V 2 µV/V	Voltage Reference Standard and Multiproduct Calibrator
DC Voltage Measure High Voltage[3]	(1 to 10) kV (10 to 20) kV (20 to 70) kV	0.9 mV/V 0.8 mV/V 0.7 mV/V	High Voltage Meter
DC Power Source[1]	0.1 W to 11 kW	2 mW/W	Multi-Function Calibrator
DC Current Source[1]	Up to 220 µA (0.22 to 2.2) mA (2.2 to 22) mA (22 to 220) mA (0.22 to 2.2) A	40 µA/A + 6 nA 35 µA/A + 7 nA 35 µA/A + 40 nA 45 µA/A + 0.7 µA 68 µA/A + 12 µA	Multiproduct Calibrator
DC High Current Source[1]	(2.2 to 11) A	395 µA/A + 480 µA	Multiproduct Calibrator and Amplifier
	(10 to 20) A (20 to 100) A	0.3 mA/A + 3 mA 0.5 mA/A + 30 mA	
	(10 to 20) A (20 to 100) A (100 to 1 000) A	3 mA/A + 2 mA 3 mA/A + 15 mA 3 mA/A + 50 mA	Multifunction Calibrator with Fluke Coil

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
DC Current Measure[1]	(0.1 to 100) nA (0.1 to 1) µA (1 to 10) µA (10 to 100) µA (0.1 to 1) mA (1 to 10) mA (10 to 100) mA (0.1 to 1) A (1 to 3) A (3 to 20) A (20 to 100) A	84 µA/A + 40 pA 13 µA/A + 40 pA 12 µA/A + 70 pA 11 µA/A + 0.6 nA 11 µA/A + 4 nA 11 µA/A + 40 nA 28 µA/A + 0.4 µA 0.11 mA/A + 10 µA 44 µA/A 27 µA/A 0.8 mA/A	Multimeter with Current Shunt
Inductance Source[1] 100 Hz to 10 kHz	(1 to 10) mH (10 to 100) mH 100 mH to 10 H 100 µH 1 mH 10 mH 100 mH 1 H 10 H	21 mH/H 21 mH/H 9 mH/H 0.25 mH/H 0.18 mH/H 0.3 mH/H 0.18 mH/H 0.23 mH/H 0.24 mH/H	Standard Inductors
Inductance Measure[1] 12 Hz to 100 kHz	100 µH to 10 H	0.4 mH/H	RLC Digibridge
Resistance Source[1] Fixed Points	0.001 Ω 0.01 Ω 0.1 Ω 1 Ω 10 Ω 100 Ω 1 kΩ 10 kΩ 100 kΩ (1 to 11) MΩ (10 to 110) MΩ (0.1 to 1) GΩ	23 µΩ/Ω 12 µΩ/Ω 3 µΩ/Ω 6 µΩ/Ω 6 µΩ/Ω 6 µΩ/Ω 13 µΩ/Ω 4 µΩ/Ω 12 µΩ/Ω 5 µΩ/Ω 12 µΩ/Ω 12 µΩ/Ω	Standard Resistors

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Resistance Source[1] Fixed Points	0 Ω 1.0 Ω 1.9 Ω 10 Ω 19 Ω 100 Ω 190 Ω 1 kΩ 1.9 kΩ 10 kΩ 19 kΩ 100 kΩ 190 kΩ 1 MΩ 1.9 MΩ 10 MΩ 19 MΩ 100 MΩ	49 µΩ 91 µΩ/Ω 91 µΩ/Ω 26 µΩ/Ω 26 µΩ/Ω 10 µΩ/Ω 10 µΩ/Ω 9 µΩ/Ω 8 µΩ/Ω 9 µΩ/Ω 8 µΩ/Ω 10 µΩ/Ω 10 µΩ/Ω 17 µΩ/Ω 18 µΩ/Ω 36 µΩ/Ω 44 µΩ/Ω 108 µΩ/Ω	Multiproduct Calibrator
Resistance Measure[1]	(0 to 10) Ω (10 to 100) Ω 100 Ω to 1 kΩ (1 to 10) kΩ (10 to 100) kΩ 100 kΩ to 1 MΩ (1 to 10) MΩ (10 to 100) MΩ 100 MΩ to 1 GΩ	6 µΩ/Ω + 30 µΩ 4 µΩ/Ω + 0.3 mΩ 2 µΩ/Ω + 0.2 mΩ 2 µΩ/Ω + 2 mΩ 3 µΩ/Ω + 20 mΩ 12 µΩ/Ω + 1 Ω 58 µΩ/Ω + 50 Ω 567 µΩ/Ω + 1 kΩ 6 mΩ/Ω + 10 kΩ	Multimeter
	(1 to 10) MΩ (10 to 100) MΩ 100 MΩ to 1 GΩ (1 to 10) GΩ (10 to 100) GΩ 100 GΩ to 1 TΩ (1 to 10) TΩ (10 to 100) TΩ 100 TΩ to 1 PΩ (1 to 10) PΩ (10 to 100) PΩ	3.4 mΩ/Ω 0.12 mΩ/Ω 0.17 mΩ/Ω 0.79 mΩ/Ω 0.56 mΩ/Ω 0.55 mΩ/Ω 2.3 mΩ/Ω 0.65 mΩ/Ω 0.56 mΩ/Ω 11 mΩ/Ω 11 mΩ/Ω	Teraohmmeter
AC Resistance Source and Measure[1]	0.001 Ω (0.1 to 1) Ω 10 Ω to 100 kΩ	0.13 mΩ 1.5 mΩ/Ω 0.46 mΩ/Ω	Calibration RL Standard

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
100 Hz to 100 kHz	1 mΩ to 0.2 Ω 0.2 Ω to 100 kΩ 100 kΩ to 10 MΩ	6.9 mΩ/Ω 0.27 mΩ/Ω 3 mΩ/Ω	RLC Digibridge and Calibration RL Standard
Capacitance Source[1] 50 Hz to 1 kHz	(0.19 to 1.1) nF (1.1 nF to 3.3) nF (3.3 to 11) nF (11 to 110) nF (110 to 330) nF (0.33 to 1.1) µF (1.1 to 3.3) µF (3.3 to 11) µF (11 to 33) µF (33 to 110) µF (110 to 330) µF (0.33 to 1.1) mF (1.1 to 3.3) mF (3.3 to 11) mF (11 to 33) mF (33 to 110) mF	4.3 mF/F + 0.01 nF 4.3 mF/F + 0.01 nF 2.2 mF/F + 0.01 nF 2.2 mF/F + 0.1 nF 2.2 mF/F + 0.3 nF 2.2 mF/F + 1 nF 2.2 mF/F + 3 nF 2.2 mF/F + 10 nF 3.4 mF/F + 30 nF 3.8 mF/F + 0.1 µF 3.8 mF/F + 0.3 µF 3.8 mF/F + 1 µF 3.8 mF/F + 3 µF 3.8 mF/F + 10 µF 8 mF/F + 30 µF 11 mF/F + 100 µF	Multifunction Calibrator
Capacitance Source[1] Fixed Values	(1, 10, 100, 1 000) pF 1 kHz to 5 MHz (5 to 13) MHz	 1.1 mF/F 1.4 mF/F	Standard Capacitors
	(10, 100, 1 000) nF 120 Hz to 100 kHz	 1.1 mF/F	
Capacitance Measure[1] 12 Hz to 100 kHz	(0.001 to 10) pF (10 to 100) pF 100 pF to 1 mF 100 pF to 1.1 µF	0.4 mF/F 0.5 mF/F 0.2 mF/F 0.6 mF/F	RLC Digibridge and Standard Capacitors
Phase Measure[1]	(10 to 60) mV 10 Hz to 10 kHz (10 to 40) kHz (40 to 100) kHz (60 to 250) mV 10 Hz to 40 kHz (40 to 100) kHz	 0.23 ° 0.4 ° 0.79 ° 0.07 ° 0.4 °	Phase Meter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Phase Measure[1]	(0.25 to 10) V 10 Hz to 40 kHz (40 to 100) kHz (10 to 120) V 10 Hz to 40 kHz (40 to 100) kHz	 0.06 ° 0.4 ° 0.08 ° 0.4 °	Phase Meter
Electrical Calibration of Thermocouple Indicators[1]	Type B (600 to 800) °C (800 to 1 000) °C (1 000 to 1 550) °C (1 550 to 1 820) °C Type C (0 to 150) °C (150 to 650) °C (650 to 1 000) °C (1 000 to 1 800) °C (1 800 to 2 316) °C Type E (-250 to -100) °C (-100 to -25) °C (-25 to 350) °C (350 to 650) °C (650 to 1 000) °C Type J (-210 to -100) °C (-100 to -30) °C (-30 to 150) °C (150 to 760) °C (760 to 1 200) °C Type K (-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 1 000) °C (1 000 to 1 372) °C Type L (-200 to -100) °C (-100 to 800) °C (800 to 900) °C	 0.48 °C 0.38 °C 0.34 °C 0.29 °C 0.26 °C 0.22 °C 0.26 °C 0.43 °C 0.71 °C 0.43 °C 0.14 °C 0.11 °C 0.14 °C 0.18 °C 0.23 °C 0.14 °C 0.11 °C 0.15 °C 0.2 °C 0.28 °C 0.16 °C 0.14 °C 0.22 °C 0.34 °C 0.42 °C 0.29 °C 0.19 °C	Multi-function Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Calibration of Thermocouple Indicators[1]	Type N (-200 to -100) °C (-100 to -25) °C (-25 to 120) °C (120 to 410) °C (410 to 1 300) °C Type R (0 to 250) °C (250 to 400) °C (400 to 1 000) °C (1 000 to 1 767) °C Type S (0 to 250) °C (250 to 1 000) °C (1 000 to 1 400) °C (1 400 to 1 767) °C Type T (-250 to -150) °C (-150 to 0) °C (0 to 120) °C (120 to 400) °C Type U (-200 to 0) °C (0 to 600) °C	 0.34 °C 0.19 °C 0.17 °C 0.16 °C 0.24 °C 0.54 °C 0.32 °C 0.29 °C 0.34 °C 0.53 °C 0.34 °C 0.32 °C 0.38 °C 0.54 °C 0.20 °C 0.14 °C 0.11 °C 0.63 °C 0.31 °C	Multifunction Calibrator
Electrical Calibration of RTD Indicators[1]	Pt 385, 100 Ω (-200 to -80) °C (-80 to 0) °C (0 to 100) °C (100 to 300) °C (300 to 400) °C (400 to 630) °C (630 to 800) °C Pt 3926, 100 Ω (-200 to 0) °C (0 to 100) °C (100 to 400) °C (400 to 630) °C	 0.05 °C 0.06 °C 0.08 °C 0.09 °C 0.1 °C 0.11°C 0.24°C 0.05 °C 0.07 °C 0.1 °C 0.12 °C	Multi-function Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Electrical Calibration of RTD Indicators[1]	Pt 3916, 100 Ω (-200 to -190) °C (-190 to 0) °C (0 to 300) °C (300 to 600) °C (600 to 630) °C Pt 385, 200 Ω (-200 to 100) °C (100 to 260) °C (260 to 600) °C (600 to 630) °C Pt 385, 500 Ω (-200 to 100) °C (100 to 260) °C (260 to 600) °C (600 to 630) °C Pt 385, 1 kΩ (-200 to 0) °C (0 to 260) °C (260 to 600) °C (600 to 630) °C PtNi 385, 100 Ω (-80 to 100) °C (100 to 260) °C Cu 427, 10 Ω (-100 to 260) °C	 0.25 °C 0.05 °C 0.08 °C 0.1 °C 0.23 °C 0.04 °C 0.05 °C 0.14 °C 0.16 °C 0.05 °C 0.06 °C 0.09 °C 0.11 °C 0.03 °C 0.05 °C 0.07 °C 0.23 °C 0.08 °C 0.14 °C 0.3 °C	Multi-function Calibrator
AC Voltage Source[1]	220 µV to 2.2 mV (10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	 0.42 mV/V + 4 µV 0.3 mV/V + 4 µV 0.3 mV/V + 4 µV 0.35 mV/V + 4 µV 0.6 mV/V + 5 µV 1.1 mV/V + 10 µV 1.5 mV/V + 20 µV 3.3 mV/V + 20 µV	Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Source[1]	(2.2 to 22) mV		Multiproduct Calibrator
	(10 to 20) Hz	0.26 mV/V + 4 µV	
	(20 to 40) Hz	0.12 mV/V + 4 µV	
	40 Hz to 20 kHz	0.1 mV/V + 4 µV	
	(20 to 50) kHz	0.21 mV/V + 4 µV	
	(50 to 100) kHz	0.53 mV/V + 5 µV	
	(100 to 300) kHz	1.1 mV/V + 10 µV	
	(300 to 500) kHz	1.4 mV/V + 20 µV	
	500 kHz to 1 MHz	2.9 mV/V + 20 µV	
	(22 to 220) mV		
	(10 to 20) Hz	0.25 mV/V + 12 µV	
	(20 to 40) Hz	0.1 mV/V + 7 µV	
	40 Hz to 20 kHz	0.1 mV/V + 7 µV	
	(20 to 50) kHz	0.21 mV/V + 7 µV	
	(50 to 100) kHz	0.48 mV/V + 17 µV	
	(100 to 300) kHz	0.86 mV/V + 20 µV	
	(300 to 500) kHz	1.4 mV/V + 25 µV	
	500 kHz to 1 MHz	2.8 mV/V + 45 µV	
	220 mV to 2.2 V		
	(10 to 20) Hz	0.25 mV/V + 40 µV	
	(20 to 40) Hz	0.1 mV/V + 15 µV	
	40 Hz to 20 kHz	0.05 mV/V + 8 µV	
	(20 to 50) kHz	0.1 mV/V + 10 µV	
	(50 to 100) kHz	0.12 mV/V + 30 µV	
	(100 to 300) kHz	0.4 mV/V + 80 µV	
	(300 to 500) kHz	1 mV/V + 0.2 mV	
	500 kHz to 1 MHz	1.7 mV/V + 0.3 mV	
	(2.2 to 22) V		
	(10 to 20) Hz	0.25 mV/V + 400 µV	
	(20 to 40) Hz	0.1 mV/V + 150 µV	
	40 Hz to 20 kHz	0.046 mV/V + 50 µV	
	(20 to 50) kHz	0.08 mV/V + 100 µV	
	(50 to 100) kHz	0.11 mV/V + 200 µV	
	(100 to 300) kHz	0.3 mV/V + 0.6 mV	
	(300 to 500) kHz	1 mV/V + 2 mV	
	500 kHz to 1 MHz	1.5 mV/V + 3.2 mV	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Source[1]	(22 to 220) V (10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	 0.25 mV/V + 4 mV 0.1 mV/V + 1.5 mV 0.055 mV/V + 0.6 mV 0.086 mV/V + 1 mV 0.15 mV/V + 2.5 mV 0.91 mV/V + 16 mV 4.8 mV/V + 40 mV 7.9 mV/V + 80 mV	Multiproduct Calibrator
AC Voltage Source[1]	(220 to 1 100) V (15 to 50) Hz 50 Hz to 1 kHz (220 to 750) V (1 to 50) kHz (50 to 100) kHz (750 to 1 100) V 40 Hz to 1 kHz (1 to 20) kHz (20 to 30) kHz	 0.3 mV/V + 16 mV 0.1 mV/V + 3.5 mV 0.36 mV/V + 11 mV 1.3 mV/V + 45 mV 0.08 mV/V + 4 mV 0.13 mV/V + 6 mV 0.36 mV/V + 11 mV	Multiproduct Calibrator with Amplifier
AC Voltage Measure[1]	22 mV Range (2 mV) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 800 kHz 1 MHz	 382 µV/V 387 µV/V 374 µV/V 352 µV/V 326 µV/V 323 µV/V 357 µV/V 323 µV/V 409 µV/V 526 µV/V 725 µV/V 818 µV/V 808 µV/V	AC/DC Transfer Standard and Multimeter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	22 mV Range (6 mV)		AC/DC Transfer Standard and Multimeter
	10 Hz	232 µV/V	
	20 Hz	224 µV/V	
	40 Hz	190 µV/V	
	100 Hz	165 µV/V	
	1 kHz	160 µV/V	
	10 kHz	175 µV/V	
	20 kHz	193 µV/V	
	50 kHz	225 µV/V	
	100 kHz	325 µV/V	
	300 kHz	488 µV/V	
	500 kHz	528 µV/V	
	800 kHz	565 µV/V	
	1 MHz	693 µV/V	
	22 mV Range (10 mV)		
	10 Hz	98 µV/V	
	20 Hz	71 µV/V	
	40 Hz	74 µV/V	
	100 Hz	72 µV/V	
	1 kHz	88 µV/V	
	10 kHz	82 µV/V	
	20 kHz	84 µV/V	
	50 kHz	98 µV/V	
	100 kHz	163 µV/V	
	300 kHz	235 µV/V	
	500 kHz	346 µV/V	
	800 kHz	349 µV/V	
	1 MHz	416 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	22 mV Range (20 mV)		AC/DC Transfer Standard and Multimeter
	10 Hz	85 µV/V	
	20 Hz	68 µV/V	
	40 Hz	75 µV/V	
	100 Hz	79 µV/V	
	1 kHz	77 µV/V	
	10 kHz	71 µV/V	
	20 kHz	77 µV/V	
	50 kHz	89 µV/V	
	100 kHz	170 µV/V	
	300 kHz	255 µV/V	
	500 kHz	331 µV/V	
	800 kHz	435 µV/V	
	1 MHz	427 µV/V	
	220 mV Range (20 mV)		
	10 Hz	92 µV/V	
	20 Hz	87 µV/V	
	40 Hz	83 µV/V	
	100 Hz	74 µV/V	
	1 kHz	68 µV/V	
	10 kHz	73 µV/V	
	20 kHz	71 µV/V	
	50 kHz	98 µV/V	
	100 kHz	157 µV/V	
	300 kHz	223 µV/V	
	500 kHz	280 µV/V	
	800 kHz	386 µV/V	
	1 MHz	356 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	220 mV Range (60 mV)		AC/DC Transfer Standard and Multimeter
	10 Hz	67 µV/V	
	20 Hz	38 µV/V	
	40 Hz	34 µV/V	
	100 Hz	39 µV/V	
	1 kHz	35 µV/V	
	10 kHz	30 µV/V	
	20 kHz	32 µV/V	
	50 kHz	42 µV/V	
	100 kHz	81 µV/V	
	300 kHz	163 µV/V	
	500 kHz	243 µV/V	
	800 kHz	282 µV/V	
	1 MHz	340 µV/V	
	220 mV Range (100 mV)		
	10 Hz	50 µV/V	
	20 Hz	30 µV/V	
	40 Hz	18 µV/V	
	100 Hz	18 µV/V	
	1 kHz	17 µV/V	
	10 kHz	19 µV/V	
	20 kHz	19 µV/V	
	50 kHz	30 µV/V	
	100 kHz	47 µV/V	
	300 kHz	91 µV/V	
	500 kHz	142 µV/V	
	800 kHz	196 µV/V	
	1 MHz	226 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	220 mV Range (200 mV) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 800 kHz 1 MHz 700 mV Range (200 mV) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 800 kHz 1 MHz	 28 µV/V 27 µV/V 16 µV/V 16 µV/V 16 µV/V 15 µV/V 17 µV/V 28 µV/V 47 µV/V 89 µV/V 135 µV/V 184 µV/V 211 µV/V 31 µV/V 28 µV/V 16 µV/V 16 µV/V 17 µV/V 16 µV/V 16 µV/V 23 µV/V 46 µV/V 81 µV/V 118 µV/V 174 µV/V 208 µV/V	AC/DC Transfer Standard and Multimeter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	700 mV Range (600 mV)		AC/DC Transfer Standard and Multimeter
	10 Hz	32 µV/V	
	20 Hz	24 µV/V	
	40 Hz	14 µV/V	
	100 Hz	14 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	13 µV/V	
	100 kHz	18 µV/V	
	300 kHz	29 µV/V	
	500 kHz	34 µV/V	
	800 kHz	54 µV/V	
	1 MHz	73 µV/V	
	2.2 V Range (0.6 V)		
	10 Hz	27 µV/V	
	20 Hz	22 µV/V	
	40 Hz	13 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	12 µV/V	
	20 kHz	13 µV/V	
	50 kHz	14 µV/V	
	100 kHz	16 µV/V	
	300 kHz	26 µV/V	
	500 kHz	29 µV/V	
	800 kHz	33 µV/V	
	1 MHz	45 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	2.2 V Range (1 V) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 800 kHz 1 MHz 2.2 V Range (2 V) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 800 kHz 1 MHz	 63 µV/V 60 µV/V 58 µV/V 58 µV/V 58 µV/V 58 µV/V 58 µV/V 58 µV/V 59 µV/V 61 µV/V 63 µV/V 65 µV/V 70 µV/V 39 µV/V 34 µV/V 31 µV/V 31 µV/V 31 µV/V 31 µV/V 31 µV/V 31 µV/V 32 µV/V 36 µV/V 39 µV/V 42 µV/V 50 µV/V	AC/DC Transfer Standard and Multimeter

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	7 V Range (2 V)		AC/DC Transfer Standard and Multimeter
	10 Hz	39 µV/V	
	20 Hz	34 µV/V	
	40 Hz	31 µV/V	
	100 Hz	31 µV/V	
	1 kHz	31 µV/V	
	10 kHz	31 µV/V	
	20 kHz	31 µV/V	
	50 kHz	31 µV/V	
	100 kHz	32 µV/V	
	300 kHz	36 µV/V	
	500 kHz	39 µV/V	
	800 kHz	42 µV/V	
	1 MHz	50 µV/V	
	7 V Range (6 V)		
	10 Hz	29 µV/V	
	20 Hz	21 µV/V	
	40 Hz	16 µV/V	
	100 Hz	16 µV/V	
	1 kHz	16 µV/V	
	10 kHz	16 µV/V	
	20 kHz	16 µV/V	
	50 kHz	16 µV/V	
	100 kHz	16 µV/V	
	300 kHz	25 µV/V	
	500 kHz	29 µV/V	
	800 kHz	33 µV/V	
	1 MHz	51 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	22 V Range (6 V)		AC/DC Transfer Standard and Multimeter
	10 Hz	29 µV/V	
	20 Hz	21 µV/V	
	40 Hz	16 µV/V	
	100 Hz	16 µV/V	
	1 kHz	16 µV/V	
	10 kHz	16 µV/V	
	20 kHz	16 µV/V	
	50 kHz	16 µV/V	
	100 kHz	16 µV/V	
	300 kHz	25 µV/V	
	500 kHz	29 µV/V	
	800 kHz	33 µV/V	
	1 MHz	51 µV/V	
	22 V Range (10 V)		
	10 Hz	32 µV/V	
	20 Hz	19 µV/V	
	40 Hz	14 µV/V	
	100 Hz	14 µV/V	
	1 kHz	14 µV/V	
	10 kHz	14 µV/V	
	20 kHz	14 µV/V	
	50 kHz	14 µV/V	
	100 kHz	15 µV/V	
	300 kHz	29 µV/V	
	500 kHz	29 µV/V	
	800 kHz	38 µV/V	
	1 MHz	46 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	22 V Range (20 V)		AC/DC Transfer Standard and Multimeter
	10 Hz	33 µV/V	
	20 Hz	20 µV/V	
	40 Hz	13 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	14 µV/V	
	100 kHz	18 µV/V	
	300 kHz	26 µV/V	
	500 kHz	28 µV/V	
	800 kHz	39 µV/V	
	1 MHz	44 µV/V	
	70 V Range (20 V)		
	10 Hz	32 µV/V	
	20 Hz	21 µV/V	
	40 Hz	13 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	14 µV/V	
	100 kHz	17 µV/V	
	300 kHz	29 µV/V	
	70 V Range (60 V)		
	10 Hz	33 µV/V	
	20 Hz	22 µV/V	
	40 Hz	13 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	15 µV/V	
	100 kHz	16 µV/V	
	300 kHz	11 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	220 V Range (60 V)		AC/DC Transfer Standard and Multimeter
	10 Hz	30 µV/V	
	20 Hz	20 µV/V	
	40 Hz	13 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	14 µV/V	
	100 kHz	16 µV/V	
	300 kHz	36 µV/V	
	220 V Range (100 V)		
	10 Hz	32 µV/V	
	20 Hz	20 µV/V	
	40 Hz	13 µV/V	
	100 Hz	14 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	14 µV/V	
	100 kHz	20 µV/V	
	220 V Range (200 V)		
	10 Hz	38 µV/V	
	20 Hz	23 µV/V	
	40 Hz	14 µV/V	
	100 Hz	14 µV/V	
	1 kHz	15 µV/V	
	10 kHz	14 µV/V	
	20 kHz	14 µV/V	
	50 kHz	16 µV/V	
	100 kHz	21 µV/V	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1]	1 000 V Range (200 V) 10 Hz 20 Hz 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 1 000 V Range (600 V) 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 1 000 V Range (1 000 V) 40 Hz 100 Hz 1 kHz 10 kHz 20 kHz 30 kHz	 41 µV/V 20 µV/V 14 µV/V 14 µV/V 15 µV/V 14 µV/V 15 µV/V 17 µV/V 33 µV/V 17 µV/V 16 µV/V 16 µV/V 18 µV/V 18 µV/V 22 µV/V 52 µV/V 16 µV/V 17 µV/V 17 µV/V 18 µV/V 16 µV/V 17 µV/V	AC/DC Transfer Standard and Multimeter
AC Voltage Measure[1] Flatness 500 kHz to 30 MHz (Relative to 1 kHz)	600 µV to 2.2 mV 500 kHz to 1.2 MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz (2.2 to 7) mV 500 kHz to 1.2 MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	 0.9 mV/V + 1 µV 0.9 mV/V + 1 µV 2 mV/V + 1 µV 3.5 mV/V + 1 µV 8.1 mV/V + 2 µV 0.9 mV/V + 1 µV 0.9 mV/V + 1 µV 1.2 mV/V + 1 µV 2.1 mV/V + 1 µV 4.3 mV/V + 1 µV	AC Measurement Standard

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Voltage Measure[1] Flatness 500 kHz to 30 MHz (Relative to 1 kHz)	(7 to 22) mV 500 kHz to 1.2 MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz 22mV to 7 V 500 kHz to 1.2 MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	 0.8 mV/V 0.8 mV/V 1.2 mV/V 2 mV/V 4.3 mV/V 0.6 mV/V 0.6 mV/V 1.2 mV/V 1.8 mV/V 4.1 mV/V	AC Measurement Standard
AC High Voltage Measure[1]	(1 to 10) kV (50 to 60) Hz (10 to 20) kV (50 to 60) Hz (20 to 50) kV (50 to 60) Hz	 0.6 mV/V 1.5 mV/V 1.4 mV/V	High Voltage Meter
AC Power Source[1]	0.1 W to 11 kW (45 to 65) Hz	2.5 mW/W	Multi-Function Calibrator
AC Current Source[1]	(9 to 220) µA (10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz 220 µA to 2.2 mA (10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (2.2 to 22) mA (10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	 0.27 mA/A + 16 nA 0.17 mA/A + 10 nA 0.14 mA/A + 8 nA 0.3 mA/A + 12 nA 1.2 mA/A + 65 nA 0.27 mA/A + 40 nA 0.17 mA/A + 35 nA 0.14 mA/A + 35 nA 0.21 mA/A + 0.11 µA 1.2 mA/A + 0.65 µA 0.27 mA/A + 0.4 µA 0.17 mA/A + 0.35 µA 0.14 mA/A + 0.35 µA 0.21 mA/A + 0.55 µA 1.2 mA/A + 5 µA	Multiproduct Calibrator

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Current Source[1]	(22 to 220) mA (10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz 220 mA to 2.2 A 20 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	 0.27 mA/A + 4 µA 0.17 mA/A + 3.5 µA 0.14 mA/A + 2.5 µA 0.21 mA/A + 3.5 µA 1.2 mA/A + 10 µA 0.28 mA/A + 35 µA 0.45 mA/A + 80 µA 6.8 mA/A + 0.16 mA	Multiproduct Calibrator
	(2.2 to 11) A 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz (2.2 to 20.5) A 50 Hz to 1 kHz (1 to 5) kHz	 0.58 mA/A + 0.17 mA 1 mA/A + 0.38 mA 3.8 mA/A + 0.75 mA 0.51 mA/A + 1 mA 0.5 mA/A + 5.8 mA	Multiproduct Calibrator and Amplifier
AC Current Source[1]	(10 to 20) A (50 to 100) Hz (100 to 400) Hz 400 Hz to 1 kHz (20 to 100) A (50 to 100) Hz (100 to 400) Hz 400 Hz to 1 kHz	 1.2 mA/A + 30 mA 2.3 mA/A + 40 mA 3.4 mA/A + 60 mA 2 mA/A + 0.15 A 3 mA/A + 0.2 A 4 mA/A + 0.3 A	Multiproduct Calibrator and Amplifier
	(10 to 20) A (45 to 65) Hz (65 to 440) Hz (20 to 100) A (45 to 65) Hz (65 to 440) Hz (100 to 1 000) A (45 to 65) Hz (65 to 440) Hz	 3 mA/A + 3 mA 9 mA/A + 3 mA 3 mA/A + 25 mA 9 mA/A + 27 mA 4 mA/A + 0.09 A 10 mA/A + 0.1 A	Multi-function Calibrator and Fluke Coil

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Current Measure[1]	(5 to 100) µA		Multimeter
	(10 to 20) Hz	4.5 mA/A + 0.03 µA	
	(20 to 45) Hz	1.7 mA/A + 0.03 µA	
	(45 to 100) Hz	0.7 mA/A + 0.03 µA	
	100 Hz to 5 kHz	0.7 mA/A + 0.03 µA	
	100 µA to 1 mA		
	(10 to 20) Hz	4.5 mA/A + 0.2 µA	
	(20 to 45) Hz	1.7 mA/A + 0.2 µA	
	(45 to 100) Hz	0.7 mA/A + 0.2 µA	
	100 Hz to 5 kHz	0.3 mA/A + 0.2 µA	
	(5 to 20) kHz	0.7 mA/A + 0.2 µA	
	(20 to 50) kHz	4.5 mA/A + 0.4 µA	
	(50 to 100) kHz	6.2 mA/A + 1.5 µA	
	(1 to 10) mA		
	(10 to 20) Hz	4.5 mA/A + 2 µA	
	(20 to 45) Hz	1.7 mA/A + 2 µA	
	(45 to 100) Hz	0.7 mA/A + 2 µA	
	100 Hz to 5 kHz	0.3 mA/A + 2 µA	
	(5 to 20) kHz	0.7 mA/A + 2 µA	
	(20 to 50) kHz	4.5 mA/A + 4 µA	
	(50 to 100) kHz	6.2 mA/A + 15 µA	
	(10 to 100) mA		
	(10 to 20) Hz	4.5 mA/A + 20 µA	
	(20 to 45) Hz	1.7 mA/A + 20 µA	
	(45 to 100) Hz	0.7 mA/A + 20 µA	
	100 Hz to 5 kHz	0.3 mA/A + 20 µA	
	(5 to 20) kHz	0.7 mA/A + 20 µA	
	(20 to 50) kHz	4.5 mA/A + 40 µA	
	(50 to 100) kHz	6.2 mA/A + 150 µA	
	100 mA to 1 A		
	(10 to 20) Hz	4.5 mA/A + 0.2 mA	
	(20 to 45) Hz	1.8 mA/A + 0.2 mA	
	(45 to 100) Hz	0.9 mA/A + 0.2 mA	
	100 Hz to 5) kHz	1.1 mA/A + 0.2 mA	
	(5 to 20) kHz	3.4 mA/A + 0.2 mA	
	(20 to 50) kHz	11 mA/A + 0.4 mA	

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Current Measure[1]	(10 to 30) mA (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (50 to 100) kHz (30 to 300) mA (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (50 to 100) kHz 300 mA to 1 A (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (50 to 100) kHz (1 to 3) A (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (50 to 100) kHz (3 to 5) A (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (50 to 100) kHz	 35 µA/A 25 µA/A 30 µA/A 60 µA/A 45 µA/A 65 µA/A 40 µA/A 30 µA/A 35 µA/A 70 µA/A 55 µA/A 85 µA/A 45 µA/A 30 µA/A 50 µA/A 0.11 mA/A 0.11 mA/A 0.2 mA/A 65 µA/A 40 µA/A 60 µA/A 0.12 mA/A 0.12 mA/A 0.2 mA/A 75 µA/A 45 µA/A 70 µA/A 0.17 mA/A 0.17 mA/A 0.31 mA/A	Multimeter and Current Shunt

Electrical – DC/Low Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
AC Current Measure[1]	(5 to 10) A (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz (10 to 20) A (10 to 20) Hz 20 Hz to 10 kHz (10 to 20) kHz (20 to 30) kHz (30 to 50) kHz	 85 µA/A 50 µA/A 80 µA/A 0.13 mA/A 0.13 mA/A 0.13 mA/A 0.07 mA/A 0.1 mA/A 0.16 mA/A 0.15 mA/A	Multimeter and Current Shunt
Oscilloscopes[1] Rise/Fall Time (10 to 90) % Sq Wave 50 Ω or 1 MΩ Load impedance – <10 kHz Horizontal Deflection Vertical Deflection	 500 ns to 100 ms 40 µVpp to 1mVpp 1 mVpp to 5 Vpp 180.10 ps to 55.00 s ±1 mV to 200 V	 1.6 ps 2.5 % of reading 0.13 % of reading 0.4 parts in 10^{-6} s 0.05 % of reading	Oscilloscope Calibrator, Active Head and Multimeter
Oscilloscopes[1] Bandwidth	100 mHz to 300 MHz (300 to 500) MHz 550 MHz to 3 GHz (3 to 26.5) GHz (26.5 to 50) GHz	2.4 % of reading 3 % of reading 4.1 % of reading 5.8 % of reading 7 % of reading	Oscilloscope Calibrator, Active Head and RF Power Sensor

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Pulse Measure[1] RMS Jitter – Period, Delay and Width	33 MHz to 3.0 GHz	1.4 % of reading + 5 ps	Wideband Oscilloscope

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
RF Power Sensors Calibration Factors[1]	(9 to 300) kHz 300 kHz to 50 MHz 50 MHz to 5 GHz (5 to 10) GHz (10 to 18) GHz (18 to 26.5) GHz (26 to 27) GHz (27 to 30) GHz (30 to 34) GHz (34 to 40) GHz (40 to 42) GHz (42 to 48) GHz (48 to 50) GHz	1 % CF 0.94 % CF 0.55 % CF 0.57 % CF 0.6 % CF 1.6 % CF 1.9 % CF 2 % CF 1.9 % CF 2 % CF 2.6 % CF 2.9 % CF 3.4 % CF	Coaxial Thermistor and Thermocouple Power Sensors
RF Power Power Meter Reference[1]	1 mW 50 MHz	0.7 % of Reading	Power Meter and Thermistor Mount
Tuned RF Power, Relative Measure[1] 150 kHz to 26.5 GHz	(20.0 to -30) dB (-30 to -130) dB	0.15 dB 0.2 dB	Measuring Receiver
RF Power Absolute Measure[1,2]	-30 dBm ≤ P < 20 dBm 50 GHz ≤ f < 60 GHz 60 GHz ≤ f < 70 GHz 75 GHz ≤ f < 75 GHz -30 dBm ≤ P < 20 dBm 75 GHz ≤ f < 76 GHz 76 GHz ≤ f < 80 GHz 80 GHz ≤ f < 90 GHz 90 GHz ≤ f < 100 GHz 100 GHz ≤ f < 110 GHz	 0.22 dB 0.21 dB 0.21 dB 0.25 dB 0.23 dB 0.22 dB 0.21 dB 0.24 dB	Power Sensor, Power Meter and Measuring Receiver
Frequency Modulation Rejection Measure[1] 150 kHz to 10 MHz 10 MHz to 26.5 GHz	Rate: 400 HZ to 1 kHz Dev.: ≤ 5 kHz peak Dev.: ≤ 50 kHz peak	 0.14 % Deviation 0.14 % Deviation	Measuring Receiver
Residuals AM[1] (rms)	150 kHz to 18 GHz (18 to 26.5) GHz	0.004 % of reading 0.006 % of reading	Measuring Receiver
Frequency Modulation Distortion Measure[1] Bandwidth: 20 Hz to 250 kHz	150 kHz to 26.5 GHz	0.15 % of reading	Measuring Receiver

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Frequency Modulation Deviation Accuracy Measure[1] 250 kHz to 10 MHz 10 MHz to 6.6 GHz (6.6 to 13.2) GHz (13.2 to 26.5) GHz	Rate: 20 Hz to 10 kHz Dev: ≤ 40 kHz Rate: 50 Hz to 200 kHz Dev: ≤ 400 kHz Rate: 50 Hz to 200 kHz Dev: ≤ 400 kHz Rate: 50 Hz to 200 kHz Dev: ≤ 400 kHz	0.1 % Modulation 0.2 % Modulation 0.3 % Modulation 0.4% Modulation	Measuring Receiver
Amplitude Modulation Rejection[3] – Measure 150 kHz to 6.6 GHz (6.6 to 26.5) GHz	Rate: 400 Hz to 1 kHz Depth.: ≤ 50 %	11 Hz 23 Hz	Measuring Receiver
Amplitude Modulation Depth Accuracy Measure[1] 150 kHz to 10 MHz 10 MHz to 3 GHz 10 MHz to 3 GHz (3 to 26.5) GHz (3 to 26.5) GHz	Rate: 50 Hz to 10 kHz Depth: (5 to 99) % Rate: 50 Hz to 100 kHz Depth: (20 to 99) % Rate: 50 Hz to 100 kHz Depth: (5 to 20) % Rate: 50 Hz to 100 kHz Depth: (20 to 99) % Rate: 50 Hz to 100 kHz Depth: (5 to 20) %	0.2 % Modulation 0.2 % Modulation 0.3 % Modulation 0.2 % Modulation 0.5 % Modulation	Measuring Receiver
Residuals FM[1] Bandwidth: 50 Hz to 3 kHz (rms)	150 kHz to 3 GHz (3 to 6.6) GHz (6.6 to 13.2) GHz (13.2 to 26.5) GHz	1.3 Hz 1.8 Hz 2.6 Hz 3.7 Hz	Measuring Receiver
Distortion Measure[1] 20 Hz to 20 kHz (20 to 100) kHz	(0 to 99.9) dB (50 to 500) kHz (50 to 500) kHz	1.3 dB 2.4 dB	Audio Analyzer
Amplitude Modulation Distortion[1] -Measure 150 kHz to 18 GHz (18 to 26.5) GHz (18 to 26.5) GHz	Depth (5 to 99) % (5 to 50) % (50 to 95) %	0.14 % Depth 0.4 % Depth 0.6 % Depth	Measuring Receiver

Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Amplitude Modulation Flatness[1] 10 MHz to 26.5 GHz	Rate: (9 Hz to 10 kHz) Depth: (5 to 99) %	0.15 % Depth	Measuring Receiver
Phase Modulation Measure[1] 100 kHz to 6.6 GHz 100 kHz to 6.6 GHz (6.6 to 13.2) GHz (6.6 to 13.2) GHz (13.2 to 26.5) GHz (13.2 to 26.5) GHz	Rate: 20 Hz to 200 kHz Deviation: > 0.7 Rad Deviation: > 0.3 Rad Rate: 20 Hz to 200 kHz Deviation: > 2 Rad Deviation: > 0.6 Rad Rate: 20 Hz to 200 kHz Deviation: > 4 Rad Deviation: > 1.2 Rad	0.1 % Modulation 0.2 % Modulation 0.1% Modulation 0.2 % Modulation 0.1 % Modulation 0.2 % Modulation	Measuring Receiver
AM Rejection Measure[1] 100 kHz to 26.5 GHz	Rate: 50 Hz to 3 kHz Deviation: > 50 % AM at 1 kHz Rate	0.003 Rad	Measuring Receiver
Phase Residuals Measure (rms) [1] 100 kHz to 26.5 GHz	Rate: 50 Hz to 3 kHz	0.001 Rad	Measuring Receiver
Phase Distortion[1] 150 kHz to 26.5 GHz	Bandwidth: 20 Hz to 250kHz	0.008 rad	Measuring Receiver
Amplitude Modulation Source[1] 10 kHz to 13.5 MHz 10 kHz to 13.5 MHz	Rate: 50 Hz to 100 kHz Depths: 5 % to 95 % Rate: 50 Hz to 100 kHz Depths: 95 % to 99 %	0.15 % Depth 0.3 % Depth	AM/FM Test Source
Frequency Modulation Source[1] 10 kHz to 432 MHz	Rate: 20 Hz to 200 kHz Dev.: ≤ 400 kHz peak	0.3 % Modulation	AM/FM Test Source
Harmonic Measurements[1] 3 Hz to 26.5 GHz (26.5 to 50) GHz	(-90 to 0.01) dB	0.8 dB 1.4 dB	Spectrum Analyzer
Phase Noise - Measure Offset Frequency ≤ 100 kHz ≤ 1 MHz ≤ 40 MHz	(0 to 10) dB	2.3 dB 3.5 dB 4.5 dB	Phase Noise System

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Electrical – RF/Microwave

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
CISPR 16 Pulse Response Quasi-Peak Detector Response Band A Band B Band C Band D Quasi-Peak to Peak Detector Relative Response Ratio Band A Bands B, C, D Quasi-Peak to Average Detector Relative Response Ratio Band A Band B Band C, D	(0 to 7) dB (1 to 20) Hz (1 to 100) Hz (1 to 100) Hz (1 to 100) Hz (0 to 70) dB 25 Hz 100 Hz (0 to 70) dB 25 Hz 500 Hz 5 kHz	 0.26 dB 0.3 dB 0.3 dB 0.3 dB 0.31 dB 0.31 dB 0.31 dB 0.31 dB 0.31 dB	Pulse Generator
CISPR 16 Pulse Response Absolute Amplitude Band A Band B Band C Quasi-Peak to RMS Relative Band A Band B, C, D	(0 to 70) dB 25 Hz 100 Hz 100 Hz (0 to 70) dB 25 Hz 100 Hz	 0.42 dB 0.42 dB 0.55 dB 0.26 dB 0.26 dB	Power Meter and Power Sensor
RF High Power – Measure (1.5 to 1 000) MHz	(0.3 to 100) W	2.4 % of reading	High RF Power Meter and Sensor

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PARAMETER	RF Absolute Power Measure [1,2]										
REFERENCE STANDARD OR EQUIPMENT	Power Sensor, Power Meter and Measuring Receiver Frequency Ranges (uncertainties in dB)										
Frequency Range	9 kHz ≤ f < 100 kHz	100 kHz ≤ f < 10 MHz	10 MHz ≤ f < 30 MHz	30 MHz ≤ f < 500 MHz	500 MHz ≤ f < 1.2 GHz	1.2 GHz ≤ f < 2 GHz	2 GHz ≤ f < 6 GHz	6 GHz ≤ f < 8 GHz	8 GHz ≤ f < 12.4 GHz	12.4 GHz ≤ f < 14 GHz	14 GHz ≤ f < 18 GHz
-70 dBm ≤ P < -60 dBm			0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
-60 dBm ≤ P < -50 dBm	0.37	0.37	0.08	0.08	0.08	0.09	0.1	0.1	0.1	0.09	0.1
-50 dBm ≤ P < -40 dBm	0.13	0.13	0.05	0.05	0.05	0.05	0.06	0.06	0.07	0.06	0.08
-40 dBm ≤ P < -30 dBm	0.06	0.06	0.05	0.05	0.05	0.05	0.06	0.06	0.06	0.06	0.08
-30 dBm ≤ P < -20 dBm	0.11	0.11	0.07	0.07	0.07	0.07	0.07	0.07	0.08	0.08	0.09
-20 dBm ≤ P < -10 dBm	0.11	0.09	0.05	0.06	0.06	0.06	0.06	0.07	0.07	0.08	0.08
-10 dBm ≤ P < 0 dBm	0.11	0.08	0.05	0.06	0.06	0.05	0.05	0.07	0.07	0.08	0.08
0 dBm ≤ P < 2 dBm	0.04	0.06	0.05	0.06	0.06	0.06	0.06	0.07	0.07	0.08	0.08
2 dBm ≤ P < 10 dBm	0.1	0.08	0.06	0.06	0.06	0.06	0.06	0.07	0.07	0.08	0.08
10 dBm ≤ P < 15 dBm	0.1	0.1	0.06	0.06	0.06	0.06	0.06	0.07	0.07	0.08	0.08
15 dBm ≤ P < 20 dBm	0.1	0.1	0.06	0.06	0.06	0.06	0.06	0.07	0.07	0.09	0.09

PARAMETER	RF Absolute Power Measure [1,2]								
REFERENCE STANDARD OR EQUIPMENT	Power Sensor, Power Meter and Measuring Receiver Frequency Ranges (uncertainties in dB)								
Frequency Range	18 GHz ≤ f < 26.5 GHz	26.5 GHz ≤ f < 33 GHz	33 GHz ≤ f < 40 GHz	40 GHz ≤ f < 45 GHz	45 GHz ≤ f < 50 GHz	50 GHz ≤ f < 55 GHz	55 GHz ≤ f < 60 GHz	60 GHz ≤ f < 67 GHz	67 GHz ≤ f < 70 GHz
-70 dBm ≤ P < -60 dBm	0.61	0.61	0.61	0.65	0.7				
-60 dBm ≤ P < -50 dBm	0.11	0.13	0.2	0.25	0.25				
-50 dBm ≤ P < -40 dBm	0.1	0.12	0.13	0.19	0.19				
-40 dBm ≤ P < -30 dBm	0.1	0.12	0.13	0.19	0.19				
-30 dBm ≤ P < -20 dBm	0.1	0.12	0.13	0.19	0.19	0.45	0.46	0.46	0.47
-20 dBm ≤ P < -10 dBm	0.09	0.1	0.14	0.15	0.17	0.24	0.24	0.24	0.27
-10 dBm ≤ P < 0 dBm	0.09	0.1	0.14	0.15	0.16	0.24	0.24	0.24	0.26
0 dBm ≤ P < 2 dBm	0.09	0.1	0.13	0.15	0.16	0.24	0.24	0.24	0.26
2 dBm ≤ P < 10 dBm	0.1	0.1	0.13	0.15	0.16	0.24	0.24	0.24	0.26
10 dBm ≤ P < 15 dBm	0.09	0.1	0.13	0.15	0.16	0.24	0.24	0.24	0.26
15 dBm ≤ P < 20 dBm	0.09	0.1	0.13	0.15	0.16	0.24	0.24	0.24	0.27

PARAMETER	(S11 - Reflection) Magnitude Uncertainty (lin)									
REFERENCE STANDARD OR EQUIPMENT	Network Analyzer, Calibration Kit and Verification Kit									
Range	Measured Magnitude (+/- Linear)									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
5 Hz to 300 kHz	0.007	0.007	0.009	0.011	0.014	0.016	0.018	0.022	0.024	0.028
300 kHz to 2 GHz	0.006 1	0.006 2	0.006 4	0.007 2	0.008	0.009	0.012	0.014	0.015	0.016
(2 to 20) GHz	0.016	0.017	0.018	0.018	0.019	0.02	0.022	0.026	0.028	0.031
(20 to 50) GHz	0.032	0.032	0.034	0.034	0.036	0.038	0.044	0.048	0.056	0.064

PARAMETER	(S11 - Reflection) Phase Uncertainty (deg)									
REFERENCE STANDARD OR EQUIPMENT	Network Analyzer, Calibration Kit and Verification Kit									
Freq:	Measured Magnitude (+/- Degrees)									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
5 Hz to 300 kHz	3.9	2.4	1.8	1.6	1.6	1.6	1.6	1.6	1.6	1.6
300 kHz to 2 GHz	3.6	1.9	1.4	1.1	0.96	0.9	0.92	0.96	0.98	1
(2 to 20) GHz	9.2	4.6	3.3	2.4	2	1.8	1.7	1.7	1.7	1.7
(20 to 50) GHz	16	9.4	6.4	5	4.2	3.8	3.6	3.6	3.6	3.7

PARAMETER	(S21 - Transmission) Magnitude Uncertainty (lin)								
REFERENCE STANDARD OR EQUIPMENT	Network Analyzer, Calibration Kit and Verification Kit								
Range	Measured Magnitude (+/- Linear)								
	0	3	6	10	20	30	40	50	60
5 Hz to 300 kHz	0.21	0.21	0.21	0.22	0.22	0.24	0.25	0.38	0.94
300 kHz to 2 GHz	0.05	0.06	0.06	0.07	0.09	0.11	0.14	0.16	0.27
(2 to 20) GHz	0.11	0.12	0.13	0.14	0.16	0.18	0.19	0.24	0.28
(20 to 50) GHz	0.17	0.18	0.2	0.2	0.21	0.42	0.46	0.5	0.54

PARAMETER	(S21 - Transmission) Phase Uncertainty (deg)								
REFERENCE STANDARD OR EQUIPMENT	Network Analyzer, Calibration Kit and Verification Kit								
Freq:	Measured Magnitude (+/- Degrees)								
	0	3	6	10	20	30	40	50	60
5 Hz to 300 kHz	1.3	1.3	1.3	1.4	1.4	1.4	1.6	2.6	3.4
300 kHz to 2 GHz	0.32	0.4	0.46	0.52	0.72	0.8	1	1.2	1.7
(2 to 20) GHz	0.7	0.8	0.84	0.9	1	1	1.3	1.6	1.9
(20 to 50) GHz	2.3	2.5	2.6	2.8	2.9	3	3.2	3.4	3.6

Length – Dimensional metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Calipers[1,2]	Up to 6 in (6 to 12) in (12 to 24) in (24 to 48) in	(283 + 4.3*L*) µin (284 + 2.1*L*) µin (289 + 1.6*L*) µin (293 + 1*L*) µin	Gage Blocks
Height Gages[1,2]	Up to 48 in	(36 + 1.2*L*) µin	Gage Blocks
End Measuring Rods[2]	(0.05 to 12) in (12 to 72) in	(3.5 + 0.8*L*) µin (4.5 + 2.3*L*) µin	Universal Length Measuring Machine, Gage Blocks
Micrometers[1,2] Resolution: 100 µin 50 µin 10 µin	Up to 48 in	(37 + 1.4*L*) µin (27 + 1.2*L*) µin (7.5 + 1.6*L*) µin	Gage Blocks
External Threads, All English and Metric 60° Threads[2] Pitch Diameter Major Diameter	Up to 4 in	(35 + 1*L*) µin (11 + 1.1*L*) µin	Universal Length Measuring Machine, Thread Wires; Gage Blocks
Internal Threads[2] Pitch Diameter Minor Diameter	Up to 4 in	(66 + 1.2*L*) µin	Universal Length Measuring Machine, Gage Blocks, Master Thread Plug Set

Length – Dimensional metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Indicators[1] Resolution: 20 µin 50 µin 100 µin 500 µin 1 000 µin	Up to 4 in	14 µin 30 µin 60 µin 280 µin 567 µin	Gage Blocks
Pin Gages, Diameter[2]	Up to 2 in	(12 + 0.34*L*) µin	Universal Length Measuring Machine, Gage Blocks
Gage Blocks[2]	Up to 6 in (6 to 24) in	(2.1 + 0.5*L*) µin (2.8 + 1.5*L*) µin	Universal Length Measuring Machine, Gage Blocks
Cylindrical Rings[2]	Up to 1 in (1 to 14) in	(9 + 0.8*L*) µin (12 + 0.8*L*) µin	Universal Length Measuring Machine, Gage Blocks
Thread Wires[2]	(4 to 120) T.P.I.	(3 + 6*D*) µin	Universal Length Measuring Machine, Gage Blocks
Rigid Rules & Tape Measures	Up to 96 in	7 300 µin	Gage Blocks
Profile Projector/ Measuring Microscope Length[1,2]	Up to 20 in	(73 + 1.3*L*) µin	Glass Scales, Gage Blocks
Glass Scales[2]	Up to 12 in	(133 + 0.08*L*) µin	Vision System
Master Levels	(4 to 18) in	24 µin/in	Surface Plate, Gage Blocks, Sine Plate
Inspection Fixtures[2] Riser Blocks, Angle Blocks, Straight Edges, Bar Parallels, V-Blocks	Up to 24 in Up to 72 in	(15 + 1.5*L*) µin (25 + 1.5*L*) µin	Universal Length Measuring Machine, Electric Comparator, Gage Blocks, Granite Surface Plate, Profile Projector, Hand Tools, Inclinometer, Cylindrical Square
Surface Plates1 – Repeat Reading (local area flatness) Overall Flatness	(6 to 120) in diagonal (12 x 12 to 72 x 44) in	30 µin 55 µin	Repeat-o-meter Electronic Level

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Length – Dimensional metrology

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Surface Roughness	≈121 µin Ra	0.88 µin	Surface Roughness Standard

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Liquid Flow Measure[1] Flow Rate – Mass DN6 to DN100 Pipe Size	15 000 kg/h	0.11 % of reading	Flow Meter
Liquid Flow – Measure[1] Flow Velocity 12.7 mm to 7.6 m Pipe Size	1 500 l/min	1.8 % of reading	Flow Meter
Piston/Plunger Operated Volumetric Apparatus (POVA) Pipettes, Syringes, Dilutors, Titrators, Piston/Displacement Burettes (Burets), Dispensers, Liquid Handlers	(0.1 to 5) µl (5 to 20) µl (20 to 50) µl (50 to 100) µl (100 to 200) µl (200 to 500) µl (0.5 to 1) ml (1 to 5) ml (5 to 10) ml	1.4 % of reading 0.65 % of reading 0.6 % of reading 0.8 % of reading 0.8 % of reading 1 % of reading 1 % of reading 1 % of reading 1 % of reading	Gravimetric Method referenced to OIML Class E1 Mass Standard and Balances
Volumetric Glassware Volumetric Apparatus, Flasks, Burettes, Volumetric Dispensers, Cylinders, Graduated Cylinders, Beakers, Vials, & Containers	(0.1 to 2) ml (2 to 25) ml (25 to 50) ml (50 to 100) ml (100 to 250) ml (250 to 500) ml (500 to 1 000) ml (1 000 to 2 000) ml	0.29 % of reading 0.12 % of reading 0.1 % of reading 0.08 % of reading 0.06 % of reading 0.04 % of reading 0.03 % of reading 0.03 % of reading	Gravimetric Method referenced to OIML Class E1 Mass Standard and Balances
Pressure[1] /Vacuum[1]	(0.2 to 1 000) psia	0.003 3 % of reading or 0.000 4 psi, whichever is greater	Deadweight Tester and Pressure Calibrator
	(0.02 to 12 140) psig	0.007 % of reading	
	(12 000 to 50 000) psig	0.2 % of reading	
Pressure[1] /Vacuum[1]	(-15 to -0.001) psig (0.001 to 300) psig	0.04 % of reading	Deadweight Tester and Pressure Calibrator

Mass and Mass Related

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Pressure[1] /Vacuum[1]	(0 to 10) Torr (0 to 100) Torr (0 to 1 000) Torr	0.02 Torr + 0.6*R* 0.097 Torr + 0.6*R* 1 Torr + 0.6*R*	Deadweight Tester and Pressure Calibrator
Air Velocity	(0.02 to 1) mph (1 to 40) mph (40 to 80) mph	8.7 % of reading 2.9 % of reading 2 % of reading	Anemometer
Scales and Balances[1,4]	(0 to 2 000) mg (0 to 20) g (0 to 200) g (0 to 1 000) g (0 to 5 000) g (0 to 20) kg (0 to 100) kg (0 to 1 000) lb	2.6 µg 20 µg 0.04 mg 0.8 mg 2 mg 4 mg 19 mg 0.32 lb	Standard Weights
Mass[2]	(1 to 500) mg (0.5 to 2) g (2 to 5) g (5 to 100) g (100 to 500) g (1 000 to 2 000) g (2 000 to 10 000) g	(0.001 2 + 0.000 003 2*W*) mg (0.001 45 + 0.005*W*) mg (0.01 + 0.000 035*W*) mg (0.018 + 0.000 04*W*) mg (0.000 2+ 0.000 2*W*) mg (0.2 + 0.000 2*W*) mg (0.2 + (0.000 5*W*) mg	Weights and Balances
	(10 to 50) kg	(0.9 + 0.02*W*) mg	
Air Flow[1]	(0.001 to 20) l/min (10 to 1 000) l/min (1 000 to 2 000) l/min	0.62 % of reading 0.65 % of reading 0.8 % of reading	Standard Flow Sensors
Torque Tools[1]	5 ozf·in to 200 lbf·in 200 lbf·in to 250 lbf·ft (250 to 1 000) lbf·ft (1 000 to 2 000) lbf·ft	0.33 % of reading 0.34 % of reading 0.34 % of reading 0.4 % of reading	Torque Calibrator
Torque Testers[1]	(15 to 200) ozf·in (4 to 150) lbf·in (2.5 to 250) lbf·ft (50 to 2 000) lbf·ft	0.12 % of reading 0.12 % of reading 0.12 % of reading 0.12 % of reading	Torque Wheel, Torque Standard, Calibration Arms, Weights
Force Compression and Tension[1]	(3 to 10 000) lbf (10 000 to 100 000) lbf (100 000 to 1 000 000) lbf	0.08 % of reading 0.1 % of reading 0.17 % of reading	Load Cell
Hydrometer[2]	(0.7 to 2) SpGr	0.1% of reading	Weighing Scale

Photometry and Radiometry

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Optical Power Measure[1]	(800 to 1 000) nm (-90 to -0.01) dB (0.01 to 10) dB (1 000 to 1 310) nm (-110 to -0.01) dB (0.01 to 10) dB (1 310 to 1 700) nm (-110 to -0.01) dB (0.01 to 10) dB	2.9 % of reading	Optical Test System
Optical Wavelength Measure[1]	(400 to 1 800) nm	0.002 nm	Wavelength Meter
Thermopile Power Sensor Measure[1]	(0.01 to 10) W	2 % of reading	Thermopile Power Sensor and Meter
Illuminance	(4 to 500) ft/candle	1.9 % of reading	Comparison to Standard Light Meter
UV Irradiance, Calibration of UV Light Meters	(0.1 to 100) mW/cm^2	4.9 % of reading	Irradiance meter, (200 to 1100) nm

Thermodynamic

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Temperature[1]	(-200 to 660) °C (660 to 1 100) °C	0.009 °C 0.75 °C	Comparison to Platinum Resistance Thermometers
Standard Platinum Resistance Thermometers Triple Point of Water Freeze Point of In Freeze Point of Zn	0.01 °C 156.5979 °C 419.5266 °C	0.000 4 °C 0.001 5 °C 0.001 6 °C	Fixed Point Cells using AC bridge
Relative Humidity[1]	(10 to 95) %RH	0.65 %RH	Humidity Generator
Dew Point[1]	(-80 to 20) °C	1.8°C	Standard Dew Point Meter
Infrared Thermometers[1]	(-20 to 1 000) °C	1.3 °C	Black Body Calibrator $\varepsilon = 0.95$, $\lambda = (8 \text{ to } 14)$ µm

Thermodynamic

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Optical Pyrometer	(600 to 900) °C (900 to 1 600) °C (1 600 to 2 600) °C (2 600 to 3 000) °C	0.6 °C 1.2 °C 1.8 °C 2.5 °C	Transfer Standard Pyrometer

Time and Frequency

Parameter/Equipment	Range	Expanded Uncertainty of Measurement (+/-)	Reference Standard, Method, and/or Equipment
Frequency Measure[1]	0.1 Hz to 225 MHz 100 MHz to 3 GHz 500 MHz to 46 GHz	6×10^{-11} Hz/Hz 6×10^{-11} Hz/Hz 6×10^{-11} Hz/Hz	GPS Receiver and Counter
Frequency Measure[1] Fixed Point	10 MHz	5×10^{-11} Hz/Hz	GPS Receiver
Frequency Source[1]	0.01 Hz to 10 MHz 10 MHz to 50.0 GHz	5×10^{-11} Hz/Hz 5×10^{-11} Hz/Hz	GPS Receiver, Signal Generator
Time Interval Measure[1]	25 ps to 1 s	1.2 % of reading + 5 ps	Wideband Oscilloscope
Photo Tachometer[1]	(0 to 100 000) rpm	0.001 6 rpm + 0.000 17 %	Generator with Lamp

Calibration and Measurement Capability (CMC) is expressed in terms of the measurement parameter, measurement range, expanded uncertainty of measurement and reference standard, method, and/or equipment. The expanded uncertainty of measurement is expressed as the standard uncertainty of the measurement multiplied by a coverage factor of 2 (k=2), corresponding to a confidence level of approximately 95%.

Notes:

1. On-site calibration service is available for this parameter, since on-site conditions are typically more variable than those in the laboratory, larger measurement uncertainties are expected on-site than what is reported on the accredited scope.
2. L = length in inches, D = diameter in inches, R = resolution of unit under test, f = frequency, P = power, W = applied weight, SpGr = specific gravity.
3. The values of the solutions reflected in this scope of accreditation are the approximate reference value at a reference temperature of 25 °C as taken from the certificate of accreditation and will vary with change in temperature.
4. The uncertainties for scales and balances are highly dependent upon the resolution of the unit under test. The uncertainty presented here does not include the resolution of the unit under test. The resolution will be included in the reported measurement uncertainty at the time of calibration
5. This scope is formatted as part of a single document including Certificate of Accreditation No. AC-1969.00

Vice President